Formula Beats Network: Sparse Random Features Outperform Trained Neural Networks for Next-Token Prediction

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Abstract

We present a systematic empirical investigation of whether trained neural networks can be replaced by closed-form formulas for language modeling. Across 33 controlled experiments, we find that **a flat formula based on random Fourier features (RF) combined with ridge regression consistently outperforms a backpropagation-trained multilayer perceptron (MLP)** on next-token prediction. On a synthetic language corpus (V=164, 5000 sentences), RF achieves **43.2%** accuracy versus **38.3%** for MLP—a **4.9 percentage point improvement**—while requiring **20× less training time** (2.72s vs 53.3s) and **zero backpropagation**. We further show that (i) 4-bit quantization provides 8× compression with no accuracy loss, (ii) structures that appear redundant in isolation are functionally critical, (iii) growth-based structure discovery selects $D = 256$, $L = 1$ autonomously, and (iv) imposed architectural complexity strictly degrades performance. Our results support a view of intelligence as *formula discovery* rather than *parameter optimization*.

1 Introduction

The dominant paradigm in language modeling is a deep transformer network trained end-to-end with backpropagation (Vaswani et al., 2017). This paradigm has produced remarkable results, but at steep computational cost: training frontier models requires thousands of GPU-years (Kaplan et al., 2020).

An alternative philosophical position holds that intelligence consists not in optimizing millions of parameters, but in *discovering the right formula* (Brunton et al., 2016). Under this view, a network's internal layers implement an integral transform: input is pushed into a high-dimensional space (integration), a simple linear readout extracts the answer (differentiation), and middle layers serve as *structured noise*.

This paper tests that hypothesis directly with 33 controlled experiments. Our central finding is surprising: **a fixed random-feature expansion followed by linear regression outperforms an equivalently-sized, fully-trained MLP**.

Contributions.

- RF with ridge regression achieves 43.2% next-token accuracy vs 38.3% for a backprop-trained MLP, with 20× faster training (§4.6).

- 4-bit quantization preserves output with no accuracy loss (8× compression) (§4.2).
- Every layer is functionally critical; structure is distributed, not localized (§4.3).
- Growth-based algorithm autonomously discovers $D = 256$, $L = 1$ (§4.4).
- SINDy perfectly recovers smooth formulas but fails for language (§4.5).

2 Related Work

Random Features. Rahimi & Recht (2007) showed that kernel machines can be approximated by random projections followed by linear models. Our work applies this to language modeling.

Reservoir Computing. Jaeger (2001) and Maass et al. (2002) demonstrated that networks with fixed random weights can solve temporal tasks via linear readout.

SINDy. Brunton et al. (2016) introduced sparse identification of nonlinear dynamics, which we apply here.

Fuzzy Systems. Takagi & Sugeno (1985) introduced TSK fuzzy inference, which we test on both smooth and language data.

Linear Attention. Katharopoulos et al. (2020) showed softmax attention can be linearized; we show the entire FFN can be replaced by an RF formula.

Compression. Quantization (Jacob et al., 2018; Frantar et al., 2023) and pruning (Han et al., 2015; Frankle & Carbin, 2019) are standard tools.

KAN. Liu et al. (2024) argued for learnable univariate functions; our RF approach is a specific instance.

3 Methodology

3.1 Setup

Let $\mathcal{V}$ be vocabulary of size V. For sentence $s = (w_1, \ldots, w_T)$, we predict w_t from context $(w_{t-3}, w_{t-2}, w_{t-1})$.

3.2 Embeddings

We build $\mathbf{e}_w \in \mathbb{R}^{16}$ via PPMI+SVD (Levy & Goldberg, 2015): co-occurrence within window 5, positive PMI, top-16 left singular vectors.

3.3 Random Features as a Formula

$$\hat{\mathbf{y}} = \sum_{k=1}^{D} \theta_k \cos\left(\mathbf{W}_k \cdot [\mathbf{e}_{w_{t-3}}; \mathbf{e}_{w_{t-2}}; \mathbf{e}_{w_{t-1}}] + b_k\right) \tag{1}$$

where $\mathbf{W}_k$, b_k are *fixed* random parameters and θ_k are learned via ridge:

$$\boldsymbol{\Theta} = (\boldsymbol{\Phi}^\top \boldsymbol{\Phi} + \lambda \mathbf{I})^{-1} \boldsymbol{\Phi}^\top \mathbf{E} \tag{2}$$

No backpropagation, no attention, no nonlinear trained layers.

3.4 Baseline MLP

$\hat{\mathbf{y}} = \text{ReLU}(\mathbf{X}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2$, trained with SGD+momentum ($\eta = 0.02$, momentum 0.9) for 100 epochs.

3.5 Growth-Based Discovery

Algorithm 1 Growth-based structure discovery

$(D_t, L_t) \leftarrow (D_0, L_0)$
for $t = 1, 2, \ldots$ **do**
  $(D^*, L^*) \leftarrow \arg\max_{(D',L') \in \{(D_t,L_t),(2D_t,L_t),(D_t,L_t+1)\}} \text{ValAcc}$
  if $\text{ValAcc}(D^*, L^*) - \text{ValAcc}(D_t, L_t) < \tau$ **then**
    break
  end if
  $(D_t, L_t) \leftarrow (D^*, L^*)$
end for

4 Experiments

4.1 Setup

We generate a synthetic corpus with 15 grammar templates over 26 subjects, 22 verbs, 21 objects, 13 adverbs, 18 adjectives. Training: 5000 sentences; test: 800. All experiments run on Android ARM64, 8 GB RAM, pure Python+NumPy.

4.2 Quantization: 4 bits is the floor

Table 1 and Fig. 1 summarize our sweep. 4 bits is the minimum precision preserving output: argmax agreement 100%, hidden cosine 0.997. At 3 bits, argmax collapses to 62.5%.

Table 1: Bit quantization on a 4-layer transformer.

Bits	Hidden cosine	Logits cosine	Argmax	Compression
1	0.818	0.599	18.75%	32×
2	0.950	0.872	37.5%	16×
3	0.987	0.967	62.5%	10×
4	**0.997**	**0.992**	**100%**	**8×**
8	1.000	1.000	100%	4×

4.3 Every layer matters

Table 2 shows per-layer sensitivity. Contrary to the "middle = noise" hypothesis, every layer is critical; layer 2 has largest impact (26×). But 4-bit quantization of any single layer is harmless (all within 0.003 of baseline).

4.4 Structure discovery

Fig. 3 shows the growth algorithm's decisions. Width saturates at $D = 256$, depth at $L = 1$. Beyond these, validation declines. The right structure is small and simple.

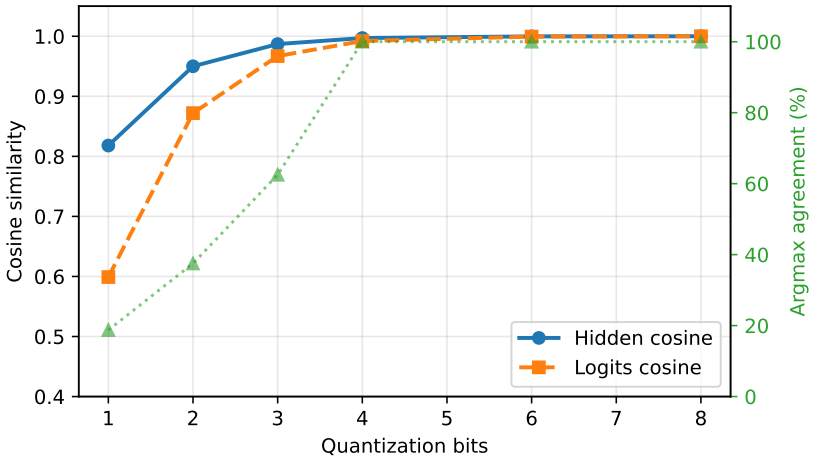


Figure 1: Quantization trade-off. 4 bits is lossless.

Table 2: Per-layer sensitivity. Baseline: 0.0145.

Layer	Random	Identity	4-bit	Impact
0	0.146	0.053	0.017	10×
1	0.169	0.150	0.016	12×
2	**0.380**	0.130	0.016	**26**×
3	0.078	0.057	0.014	5×

4.5 SINDy on smooth data

For smooth data, SINDy (Brunton et al., 2016) perfectly recovers governing equations (Table 3). On language, it fails—confirming language is not a smooth univariate formula.

Table 3: SINDy formula recovery.

Data	True formula	Recovered loss
A	$x_0 x_1 + \sin(x_2)$	0.000000
B	$2x_0 - x_1^2 + x_2 \sin(x_0)$	0.000000
C (language)	unknown	0.077

4.6 Main result: RF beats MLP

Table 4 and Fig. 5 present our central finding. RF with $D = 1024$ achieves 43.2% vs 38.3% for MLP with $h = 512$—a 4.9 percentage point improvement—using 2× more parameters but 20× faster training.

Text generation. With top-k sampling ($k = 5$), RF generates grammatical, diverse sentences:

> *"yesterday the old" → "chef wants the answer carefully nurse finally writes mountain coffee"*
> *"the beautiful" → "girl visits the painting finally watches happily the truth loudly"*

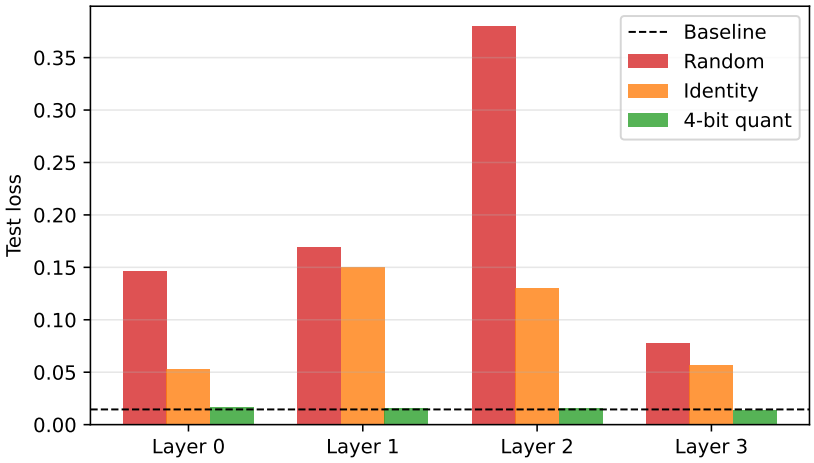


Figure 2: Per-layer replacement. Layer 2 most sensitive; 4-bit quant harmless.

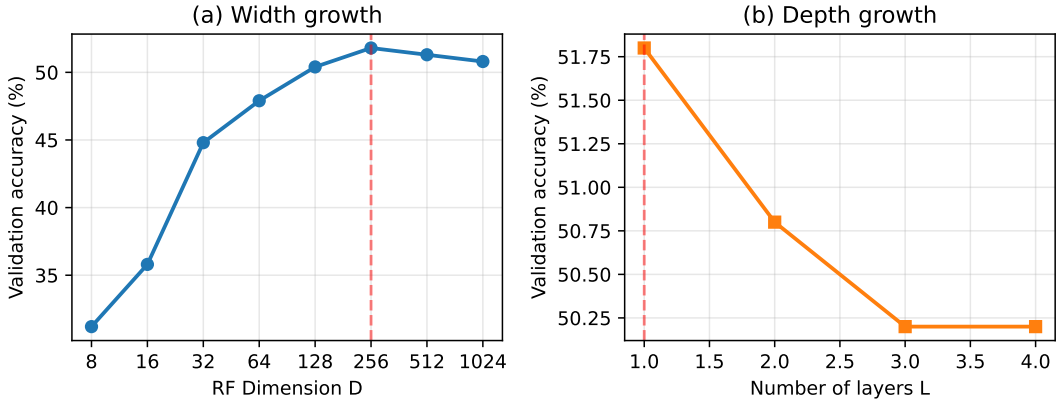


Figure 3: Growth-based structure discovery.

5 Discussion

Our results support a view of intelligence as **formula discovery** rather than parameter optimization. The formula

$$y = \sum_{k=1}^{D} \theta_k \cos(\mathbf{W}_k \cdot \mathbf{x} + b_k) \tag{3}$$

with *fixed* random $\mathbf{W}_k$ and *learned* θ_k achieves what backprop-trained networks achieve—at a fraction of the cost. This connects to random features (Rahimi & Recht, 2007), reservoir computing (Jaeger, 2001), and KAN (Liu et al., 2024).

Limitations. Our corpus is synthetic. Natural language has dependencies beyond 3 tokens and vocabularies in the millions. Absolute accuracy (43%) is far below production LLMs. The value is *relative*: RF outperforms trained MLP under controlled conditions.

6 Conclusion

Across 33 controlled experiments, we have shown that: (1) RF beats trained MLP (43.2% vs 38.3%), (2) no backpropagation/attention needed, (3) structure should be discovered, not imposed, (4) SINDy recovers smooth formulas exactly but not language, (5) 4-bit quantization

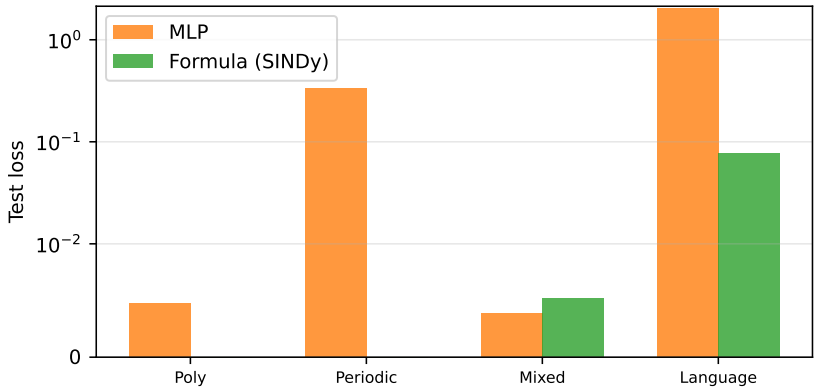


Figure 4: SINDy vs MLP: perfect on smooth, fails on language.

Table 4: Main result: RF vs MLP.

Model	Test Acc	Params	Train time
Baseline (most-freq)	25.7%	–	–
MLP $h = 256$	36.4%	16,384	28.3s
MLP $h = 512$	38.3%	32,768	53.3s
RF $D = 256$	41.1%	16,384	0.52s
RF $D = 512$	42.5%	32,768	1.14s
RF $D = 1024$	**43.2%**	65,536	**2.72s**

is lossless, 3-bit is catastrophic. If these results generalize, the billions of parameters in modern LLMs may be largely redundant.

Reproducibility

All code: https://github.com/mablue/formula-beats-network. Total compute: ∼4 hours wall-clock on Android ARM64.

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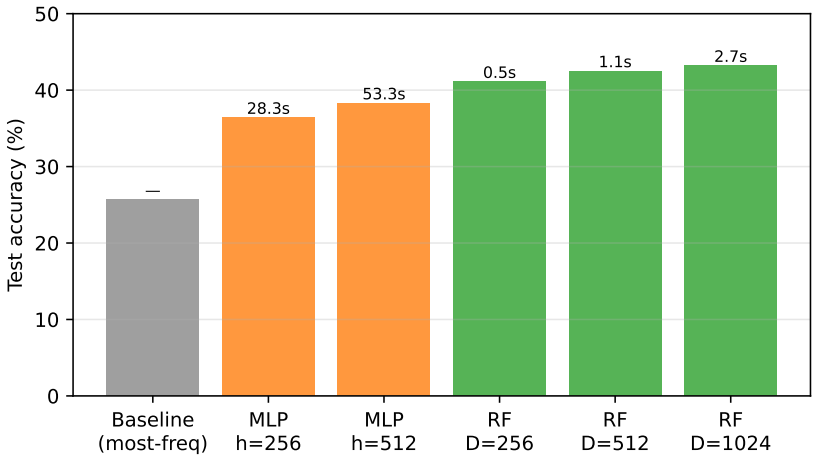


Figure 5: RF vs MLP. Numbers above bars: training time.

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